9.6 Strange Dice

WORKSHEET

The question

STEP I 1993, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q1

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself If the first die has p_i faces showing i and the second has q_j faces showing j, how many ways are there to throw a total of t?

STAGE 1 The extremes

Ask yourself There is no way to throw 2, nor 12. What does that say about the 1s and the 6s? There *are* two ways to throw 3. Which die has the 1s?

STAGE 2 Working upwards

Ask yourself Write down the equation for a total of 4, remembering $b_1 = 0$. What does it say in each of the two cases?

STAGE 3 Case (a) fails

? **Ask yourself** In case (a), write down the equation for a total of 7, and use $b_5 + b_6 = 5$.

STAGE 4 **Case (b) succeeds**

? **Ask yourself** In case (b), use the totals 6 and 11, and the fact that $a_6 b_6 = 0$.

What's really going on

Dice as polynomials

Write each die as a polynomial, with one term x^i for each face showing i. An ordinary die is $x + x^2 + x^3 + x^4 + x^5 + x^6$, and the dice in the question are

$$A(x) = x + x^4 + x^5 + 3x^6, \qquad B(x) = 2x^2 + x^3 + x^4 + 2x^5.$$

Multiplying, each pair of faces contributes $x^i x^j = x^{i+j}$, so the coefficient of x^t in $A(x)B(x)$ is the number of ways to throw a total of t. This is the generating function idea of Section 9.4 once more: the table of totals is just the product $A(x)B(x)$, and our case-by-case deduction was a way of factorising it.

Sicherman's dice

For two ordinary dice the product is $(x + x^2 + \cdots + x^6)^2$. Any pair of dice with the same totals must have polynomials $P(x)$, $Q(x)$ with $P(x)Q(x) = (x + \cdots + x^6)^2$. Now factorise:

$$x + x^2 + \cdots + x^6 = x(1 + x)(1 + x + x^2)(1 - x + x^2),$$

and these factors cannot be broken down further (into polynomials with integer coefficients). So each of P and Q must be built from the factors x, $1+x$, $1+x+x^2$, $1-x+x^2$, each appearing twice in total. Three conditions narrow the choice:

- each die has 6 faces, so $P(1) = Q(1) = 6$; the factors take the values 1, 2, 3, 1 at $x = 1$, so each die gets one $1+x$ and one $1+x+x^2$;
- no face shows 0, so $P(0) = Q(0) = 0$: each die gets one factor x;
- the only freedom left is how the two copies of $1-x+x^2$ are shared. One each gives ordinary dice. Both on one die gives

$$x(1+x)(1+x+x^2) = x+2x^2+2x^3+x^4,$$
$$x(1+x)(1+x+x^2)(1-x+x^2)^2 = x+x^3+x^4+x^5+x^6+x^8.$$

These are Sicherman's dice, $1,2,2,3,3,4$ and $1,3,4,5,6,8$ (Figure 9.6.1), and the argument proves they are the only alternative. The key fact is that polynomials, like integers, factorise uniquely into irreducibles: number theory and combinatorics meet.

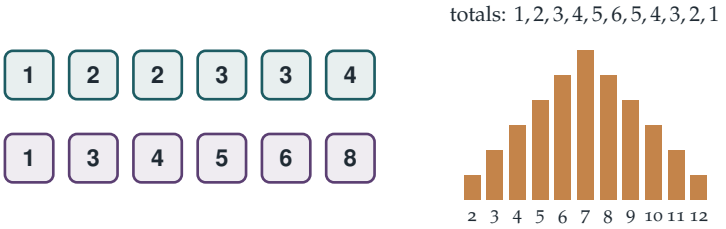


Figure 9.6.1 Sicherman's dice. Thrown together, they give every total from 2 to 12 with exactly the same frequency as two ordinary dice (right): one way to throw 2, two ways to throw 3, and so on up to six ways to throw 7.

Probability generating functions

In probability the same idea appears as the *probability generating function*: a random variable X taking values $0,1,2,\ldots$ has $G(t) = E(t^X) = \sum P(X=k)t^k$, and for independent X and Y, $E(t^{X+Y}) = E(t^X)E(t^Y)$. Sums of independent random variables correspond to products of generating functions. STEP III 2003 Q14 (in the exercises) finds Sicherman's dice this way, and Section 9.13 uses a related argument to show that no loading of two dice can make all totals equally likely.

Try this next

1 **Tetrahedral dice.** Try STEP III 2003, Question 14 (step.maths.org/questions/03-s3-q14), which asks Sicherman's question for four-faced dice. (*Hint:* $x+x^2+x^3+x^4 = x(1+x)(1+x^2)$.)

2 **Checking the table.** Multiply out $(x+x^4+x^5+3x^6)(2x^2+x^3+x^4+2x^5)$ and check it against the table in the question.

3 **Three dice.** Are there three six-sided dice, not all standard, labelled with positive integers, whose totals have the same distribution as three standard dice? (*Hint:* the factors of $(x + \cdots + x^6)^3$ now include three copies of $1 - x + x^2$.)

4 **A coin and an octahedron.** An eight-sided die labelled $1, \ldots, 8$ and a two-sided "die" (a coin) labelled $1, 2$: can they be relabelled differently with the same totals?

Further reading

- **Martin Gardner, *Penrose Tiles to Trapdoor Ciphers*** (Mathematical Association of America). Includes the column on Sicherman dice.
- **Joseph Gallian & David Rusin, "Cyclotomic polynomials and nonstandard dice"**, *Discrete Mathematics* 27 (1979). The general theory, for dice with any number of faces.
- **Herbert Wilf, *generatingfunctionology*** (free online), Chapter 1.

The full guided solution is in Section 9.6:
https://beyondstep.keithmcnulty.org/ch09/strange-dice.html